ON HIGHLY RAMSEY INFINITE GRAPHS

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Abstract. We show that, for $k \geq 3$, there exists a positive constant c such that for large enough n there are 2^{cn^2} non-isomorphic graphs on at most n vertices that are 2-ramsey-minimal for the odd cycle C_{2k+1}.

1. Introduction

A graph G is *r-ramsey* for a graph H if any r-colouring of the edges of G yields a monochromatic copy of the graph H. G is *r-ramsey-minimal* for H if it is r-ramsey for H, but none of its proper subgraphs are. H is *r-ramsey-infinite* if there is an infinite set of graphs that are r-ramsey-minimal for H.

For results about ramsey minimal and ramsey infinite graphs, see, for example, [1] - [13], and [16].

If for some graph H, there exists a constant $c > 0$ such that for any large enough n there are at least 2^{cn^2} non-isomorphic graphs on at most n vertices that are r-ramsey-minimal for H, then we say that H is *highly r-ramsey-infinite*. The word "highly" comes from the fact that up to the constant c, the above lower bound is the best possible.

Strengthening a result of [7] which states that complete graphs are 2-ramsey-infinite, it was shown in [10] that for any 3-connected graph H, and all $r \geq 3$, there is a positive constant c such that for all n large enough, there are at least $2^{cn \log n}$ non-isomorphic graphs on at most n vertices that are r-ramsey-minimal for H.

In [18] it was shown, for $k \geq 3$ and $r \geq 2$ that the complete graph K_k is highly r-ramsey-infinite.

The papers [7], [10] and [18] all used the existence of a graph called a signal-sender in their constructions.

In this paper we prove the existence of signal-senders for odd-cycles, and show that for odd $k \geq 7$, and $r \geq 2$, the cycle C_k is highly r-ramsey-infinite.

Precisely, we prove the following theorem.

Theorem 1.1. *For integer $r \geq 2$ and odd integer $k \geq 7$, there exist constants $c = c(r,k) > 0$ and $n_0 = n_0(r,k)$ such that for all $n \geq n_0$ there exist at least 2^{cn^2} non-isomorphic graphs, each on at most n vertices, that are r-ramsey-minimal for the odd cycle C_k.*

The proof will be constructive, and done in such a way that the 2^{cn^2} different graphs all have odd girth k, so are, infact, r-ramsey-minimal for induced copies of C_k. This stronger version of ramsey minimality has been investigated in such papers as [1], [8], and [15].

2. Preliminary Results

Throughout the paper, we will often identify a graph or hypergraph G with its edgeset $E(G)$, and let $[r]$ denote the set $\{1, \ldots, r\}$. Given a function f defined on a set S we will let $f(S)$ denote the set $\{f(s) \mid s \in S\}$. An r-colouring of a graph H is a mapping of the edges to the set $[r]$. An *H-free r-colouring* of a graph G is an r-colouring such that there is no monochromatic copy of H. An r-vertex-colouring of a graph G is a mapping of its vertices to the set $[r]$ such that adjacent vertices are assigned different images.

We now introduce signal senders- these are graphs that are very useful in the construction of ramsey-minimal graphs. See [7], as well as [10] and [18]. In these papers, senders have been shown to exist for any 3-connected graph H. Here, we define senders, and in Section 4 we prove their existence for cycles.

Definition 2.1. Given an integer $k \geq 4$, the *negative signal sender* $S = S^-(r, C_k, e, f)$ is a graph containing edges e and f with the following properties.

(i) S is not r-ramsey with respect to C_k.
(ii) For every C_k-free r-colouring of S, edges e and f have different colours.
(iii) S has girth k and distance at least $k+1$ between edges e and f.

A *positive signal sender* $S = S^+(r, C_k, e, f)$ is defined similarily but we replace the word 'different' in property (ii) with 'the same'. The edges e and f are referred to as the *interface edges*.

Note that the definition is usually made without property (iii), but since we we will always need this property in this paper, we include it in the definition.

In Section 4 we prove the following, using a proof similar to that in [10] and [18].

Lemma 2.2. *For $r \geq 2$ and $k \geq 4$ there exist*

a) *negative signal senders $S^-(r, C_k, e, f)$, and*
b) *positive signal senders $S^+(r, C_k, e, f)$.*

Remark 2.3. We will often use these senders in constructions in the following way. Given a graph G with edges e_1 and e_2 we will take a copy S of $S^-(r, C_k, e, f)$ or $S^+(r, C_k, e, f)$, and we will identify edges e_1 and e_2 with edges e and f, of S, respectively. When we do this we say that we "connect the edges e_1 and e_2, with S."

When we use such a process we can say the following.

Proposition 2.4. *Given a graph G with edges e_1 and e_2, when we connect the edges e_1 and e_2 with a copy S of $S^-(r, C_k, e, f)$ or $S^+(r, C_k, e, f)$, we introduce no new cycles of length k or less.*

Proof. Let C be a new cycle introduced in the construction. Then either C was a path in S, with one endpoint in e and one in f, such that these endpoints were identified in the construction, or C strictly contains a path in S whose endpoints are in $V(\{e, f\})$. Either way, property (iii) of Definiton 2.1 ensures that C has length at least $k+1$.

□

Lemma 2.2, and the following result of V. Müller [14] are the main tools that we will use in the proof of Theorem 1.1.

Lemma 2.5 ([14]). *Let $k \geq 3$ and $r \geq 3$ be integers, and $\Gamma \subseteq \{\nu \mid \nu : W \to [r]\}$ be a set of mappings from some set W to $[r]$, which is closed under permutation of $[r]$. Then, there exists a graph M*

(i) *of girth at least k,*
(ii) *with $W \subset V(M)$, and such that*
(iii) *a mapping $\nu : W \to [r]$ can be extended to an r-vertex-colouring of M if and only if ν is in Γ. (Moreover M can be assumed to be minimal with respect to this property.)*

Remark 2.6. Throughout the paper we will define maps ν on indexed sets such as $W = \{w^1, \ldots, w^r\}$. When we do this, we will assume ν to be defined on any copy $W_i = \{w_i^1, \ldots, w_i^r\}$ of W in the natural way:

$$\nu(w_i^j) = \nu(w^j).$$

Senders can be used to convert many colour-critical constructions to ramsey-minimal constructions. We will use them to translate the above result of Müller into the following version that is useful in edge-colouring applications. Moreover, we use an explicit construction to extend the result to the case $r = 2$. (Müller's result cannot be extended to the case $r = 2$.) Though we state it only for the graphs C_k that we are interested in, one could replace C_k in the following lemma with any graph H for which there exists a sender of girth k.

Lemma 2.7. *Let $k \geq 3$ and $r \geq 2$ be integers. Let $\Gamma \subseteq \{\nu \mid \nu : W \to [r]\}$ be a set of mappings, from some set W to $[r]$, which is closed under permutation of $[r]$. Then, there exists a graph M*

(i) *of girth k,*
(ii) *with $W \subset M$ $(= E(M))$,*
(iii) *with distance at least $k+1$ between edges of W, and such that*
(iv) *a mapping $\nu : W \to [r]$ can be extended to a C_k-free r-colouring of M if and only if ν is in Γ. (Moreover, M can be assumed to be minimal with respect to this property.)*

Proof. First we prove the lemma for $r > 2$, using Lemma 2.5. We will then prove the case $r = 2$ with an explicit construction.

Case $r > 2$: Given $k, r > 2, W$, and Γ, let M' be the r-chromatic graph returned by Lemma 2.5 for these values. We construct the graph M meeting properties (i)-(iv) of Lemma 2.7 as follows.

- For every vertex v in M' let e_v be an independent edge.
- For every edge uv in M' join edges e_u and e_v with a copy S_{uv} of the negative signal sender $S^- = S^-(r, C_k, e, f)$. (See Remark 2.3).

Property (i) of Lemma 2.7 comes from the fact that the senders had girth k, and from Proposition 2.4.

Property (ii) is achieved by identifying W with the edgeset $E_W = \{e_v \mid v \in W \subset V(M')\}$, derived from it.

Property (iii) comes from the fact that the interface edges of S^- had the same property.

To see property (iv), let ν be a mapping from W to $[r]$. If ν is in Γ then it is the restriction, to W, of some r-vertex-colouring χ' of M'. We define $\chi : M \to [r]$, which extends $\nu : W \to [r]$ (a mapping from $M = E(M)$ via the identification

$W = E_W$) as follows. For $v \in V(M')$ let $\chi(e_v) = \chi'(v)$, thus χ agrees with ν on $W = E_W$. Now for every edge uv in M', $\chi(e_u) = \chi'(u) \neq \chi'(v) = \chi(e_v)$, so χ can be extended, by Definition 2.1, to a C_k-free colouring of S_{uv}. Since M was constructed be joining independent edges with copies of S^-, Proposition 2.4 gives us that any copy of C_k in M is within one of the senders S_{uv}. Thus χ is a C_k-free r-colouring of M.

On the other hand, assume that $\nu \notin \Gamma$, but that it extends to a C_k-free r-colouring χ of M. Then for every edge $uv \in M'$, the properties of S_{uv} imply that $\chi(e_u) \neq \chi(e_v)$. But then $\chi' : V(M') \to [r]$ defined by $\chi'(v) = \chi(e_v)$ is a proper r-colouring of M' that extends ν. This contradicts the fact that $\nu \notin \Gamma$.

This proves the Lemma when $r > 2$.

Case $r = 2$: Let $r = 2$, and $k > 2, W$, and Γ be given. Let $\overline{\Gamma}$ be the set of 2-colourings of W not in Γ. Let $\gamma \in \overline{\Gamma}$.

Claim 2.8. *There exists a graph X_γ*

(i) *of girth k*
(ii) *with $W \subset X_\gamma$ $(= E(X_\gamma))$.*
(iii) *with distance at least $k+1$ between edges of W, and such that*
(iv) *any 2-colouring of W except γ (up to permutation of the set $[2]$) extends to a C_k-free 2-colouring χ of X_γ.*

Assuming the claim, the union of these graphs X_γ over all $\gamma \in \overline{\Gamma}$, where the only common vertices are those in the edges of W, is our desired graph M. Indeed, if W is coloured by $\nu \in \Gamma$, then $\nu \neq \gamma$ for any $\gamma \in \overline{\Gamma}$, so by property (iv) of the claim ν can be extended to each of the graphs X_γ. On the other hand, if W is coloured with any other 2-colouring, then this colouring is some $\gamma \in \overline{\Gamma}$, so by property (iv), cannot be extended to a C_k-free 2-colouring of $X_\gamma < M$. This gives us property (iv) of the lemma. The other properties required for the lemma follow immediately from the corresponding properties of the claim.

We finish the proof of the lemma by proving the claim. We do this constructively by induction on $|W|$. Our base case will be the case $|W| \leq k$. This is the only case that we use in this paper, and we give it in detail. We only sketch the induction. For it, our assumption will be that we have proved the claim, and so the lemma, for any smaller set W.

Proof of claim. Let $\gamma \in \overline{\Gamma}$ be given. Assume that $|W| \leq k$. Take W, and a copy C_γ of the graph C_k, and construct X_γ containing W and C_γ as follows.

- Let ϕ be an onto map from $C_\gamma = E(C_\gamma)$ to W.
- For every $c \in C_\gamma$, connect c to $\phi(c)$ with a copy of $S^+ = S^+(2, C_k, e, f)$ if $\gamma(\phi(c)) = 1$, or a copy of $S^- = S^-(2, C_k, e, f)$ if $\gamma(\phi(c)) = 2$.

We now show that the graph C_γ has the properties required by the claim. Property (i) comes from Proposition 2.4 the fact that the graphs C_k, S^-, and S^+ all have girth k. Property (ii) is trivial. Property (iii) comes from the fact that the edges of W were disconnected and then we connected them with senders that have distance at least $k+1$ between their interface edges. To see property (iv) observe that under any 2-colouring χ of X_γ in which the copies of S^+ and S^- are C_k-free, the subgraph C_γ is monochromatic if and only if $\chi|_W = \gamma$ up to permutation of [2]. Since apart from C_γ, the only copies of C_k in X_γ are in copies of S^+ and S^-, this completes the proof of the case $|W| \leq k$.

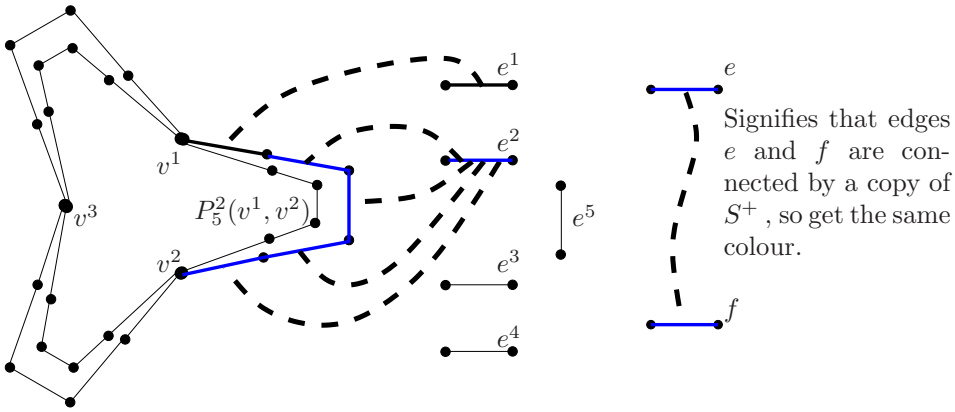

FIGURE 1. Construction 3.1: $F(2, k, 5, K_3)$. Step (ii).

We now sketch the induction on $|W|$. Assume that $|W| > k$, and that the lemma, and so the claim, is proved for any set W' with $|W'| < |W|$. Partition the edges of W into parts $W_1, \ldots, W_w$ of at most $k-1$ edges each. For each $i \in [w]$, let W_i' be $W_i \cup \{e_i\}$ for some new edge e_i, and extend γ to W_i' by setting $\gamma(e_i) = 1$. By induction we can build a graph $X_{\gamma,i}$ such that for any C_k-free colouring χ of $X_{\gamma,i}$, $\chi|_{W_i} = \gamma$ implies that $\chi(e_i) = 1$. Letting $W' = \{e_i \mid i \in [w]\}$ we can build by induction a graph X_γ' such that the only C_k-free 2-colourings restrict to non-monochromatic maps of W'. Letting X_γ be the union of X_γ' and all the $X_{i,\gamma}$, we get our desired graph. □

This completes the proof of the claim, and so of the lemma. □

3. The Proof of Theorem 1.1

The proof of the theorem will be by construction. Before we get to the main construction for the proof of the theorem, we introduce two necessary preliminary constructions.

Construction 3.1 ($F(r, k, k', H)$)**.** Given the integers $r \geq 2$, $k' \geq 2$, $k \geq 5$, and graph H, we define the graph $F = F(r, k, k', H)$ as follows. (See Figure 1)

(i) Begin with the vertex set $V(H) = \{v^1, \ldots, v^h\}$ of H and the set $E = \{e^1, \ldots, e^{2r+1}$ of $2r+1$ independent edges.

(ii) For every edge $e = v^j v^{j'}$ of H, let F contain the set

$$\mathcal{P}_{k'}(v^j, v^{j'}) = \{P_{k'}^1(v^j, v^{j'}), \ldots, P_{k'}^r(v^j, v^{j'})\},$$

of r distinct, internally vertex-disjoint paths of length k' from v^j to $v^{j'}$. (F does not contain the edge $v^j v^{j'}$ of H.)

(iii) For all $v^j v^{j'} \in E(H)$ and all $P = P_{k'}^\ell(v^j, v^{j'}) \in \mathcal{P}_{k'}(v^j, v^{j'})$ let the first edge of P be connected to $e^{2\ell-1}$ by a copy of the positive sender S^+ and let every other edge of P be connected to $e^{2\ell}$ in the same way. (Observe that the edge e^{2r+1} is an independent edge.)

Denote the set of all of paths introduced in step (ii), by $\mathcal{P}$.

We now collect some properties of the above constrution.

Proposition 3.2. *Where $F = F(r, k, k', H)$ is the graph from Construction 3.1, F has the following properties.*

(i) *If $k' \cdot g > k$, where g is the girth of H, then the only copies of C_k in F are in the copies of S^+. Thus any r-colouring of the set E can be extended to a C_k-free r-colouring of the graph F.*

(ii) *Let $\ell \in [r]$, $v^j v^{j'} \in H$, and χ be a C_k-free r-colouring of F. Then $\chi(P^\ell_{k'}(v^j, v^{j'})) = \chi(\{e^{2\ell-1}, e^{2\ell}\})$.*

Proof. It is easy to see that there are only copies of C_k in $\mathcal{P} < F$ if $k' \cdot g \leq k$. Thus property (i) follows from Proposition 2.4 and the fact that F is constructed from $\mathcal{P}$ and the independent set E by connecting edges with copies of S^+.

Property (ii) comes from the fact that under any C_k-free r-colouring of F, the interface edges e and f of any copy of S^+ introduced in step (iii) of the construction, get the same colour. □

By property (ii) of Proposition 3.2, the graph $F = F(r, k, k', H)$, has the property that the paths of $\mathcal{P}$ are all monochromatic under some C_k-free r-colouring χ of F if the $\chi(e^{2\ell-1}) = \chi(e^{2\ell})$ for all $\ell = 1, \ldots, r$.

We now construct a graph, using Lemma 2.7, that we will later use to connect several copies of F together in such a way that under any C_k-free r-colouring, all paths in $\mathcal{P}$ are monochromatic for at least one of the copies of F.

Construction 3.3. ($M(r, k)$) Given $r \geq 2$ and odd $k \geq 3$, let $W' = \{e^1, \ldots, e^{2r+1}\}$, and define the following maps from W' to $[r]$.

- $\nu_1 : e^{2\ell-1}, e^{2\ell} \mapsto \ell$ for $\ell = 1, \ldots, r$ and $e^{2r+1} \mapsto 1$.
- $\nu_2 : e^{2\ell-1} \mapsto \ell, e^{2\ell} \mapsto \ell + 1$ for $\ell = 1, \ldots, r$ and $e^{2r+1} \mapsto 1$
- $\nu_3 : e^{2\ell-1} \mapsto \ell, e^{2\ell} \mapsto \ell + 1$ for $\ell = 1, \ldots, r$, and $e^{2r+1} \mapsto 2$.

(Images are mod r). Now let $W = W_1 \cup W_2$ be the union of two copies $W_1 = \{e^1_1, \ldots, e^{2r+1}_1, \}$ and $W_2 = \{e^1_2, \ldots, e^{2r+1}_2\}$ of W'. (So the ν_i are defined on W_1 and W_2 as per Remark 2.6.) Define the map $\nu_{ij} : W \to [r]$ by $\nu_{ij}|_{W_1} = \nu_i$ and $\nu_{ij}|_{W_2} = \nu_j$. Define Γ to be the closure under permutation of $[r]$ of the set

$$\{\nu_{ij} \mid 1 \leq i \neq j \leq 3\},$$

of six maps from W to $[r]$.

Let $M(r, k)$ be the graph M returned by Lemma 2.7 for this choice of r,k, $H = C_k$, W, and Γ.

Remark 3.4. We will use the graph $M(r, k)$ in constuctions, in the following way. Given two indexed sets $F_1 = \{f^1_1, \ldots, f^{2r+1}_1\}$ and $F_2 = \{f^1_2, \ldots, f^{2r+1}_2\}$ of $2r + 1$ independent edges each, we will take a copy M of the graph $M(r, k)$. For all $\ell = 1, \ldots, 2r + 1$ we will identify e^ℓ_1 of M with f^ℓ_1 and e^ℓ_2 of M with f^ℓ_2. When we do this, we say that we "connect F_1 and F_2 with M"

As we had for the senders, we also have

Proposition 3.5. *Given a graph G with edge sets F_1 and F_2, when we connect F_1 and F_2 with a copy of $M(r, k)$ we introduce no new cycles of length k or less.*

Proof. This follows immediatly from Proposition 2.4 and the fact that $M(r, k)$ is made up entirely of independent edges that have been connected by copies of $S^-(r, C_k, e, f)$. □

We are now ready to present the construction that we use in the proof of the Theorem. Before we begin, we give a very brief sketch of the proof. For given m (which will be some constant fraction of n), let $\mathcal{Q} = \{Q_i \subset [m] \mid i = 1, \dots, m\}$ be an ordered set of m (not necessarily distinct) subsets of the set $[m]$. For each of the $2^{\binom{m}{2}} \sim 2^{m^2}$ such sets $\mathcal{Q}$ we will construct a graph $G(\mathcal{Q})$ on at most n vertices, that is r-ramsey for C_k. We will then show that for some constant $c' = c'(r,k)$ independent of m, the r-ramsey-minimal subgraphs of these graphs fall into at least $2^{c'm^2}$ distinct isomorphism classes. Since n is a constant times m this will be 2^{cn^2} classes for some constant c, independent of n.

The actual proof of Theorem 1.1 comes after the following construction and proposition. Recall that an $(r+1)$-critical graph is a graph that has no r-vertex-colouring, but for which every proper subgraph does have an r-vertex-colouring.

Construction 3.6. ($G(\mathcal{Q}) = G(\mathcal{Q}, k, r, m)$) Given integer $r \geq 2$, odd integers $k \geq 7$ and m, and $\mathcal{Q} = \{Q_i \subset [m] \mid i = 1, \dots, m\}$ as described above, define the graph $G(\mathcal{Q}) = G(\mathcal{Q}, k, r, m)$ as follows. (See Figure 2.) If $k = 9$, let H be a triangle-free $(r+1)$-critical graph, for any other k, let $H = K_{r+1}$. Let $V(H) = \{v^1, \dots, v^h\}$.

(i) For $i = \pm 1, \dots, \pm m$ let F_i^A be a copy of $F(r, k, (k-1)/2, H)$ from Construction 3.1. Let $V_i^A = \{a_i^1, \dots, a_i^h\} \subset V(F_i^A)$ be copy of the set $\{v^1, \dots, v^h\}$ in $V(F_i^A)$, let $E_i^A = \{e_i^1, \dots, e_i^{2r+1}\}$ be the copy of E in F_i^A, and let $\mathcal{P}_i^A = \{\mathcal{P}_{(k-1)/2}(a_i^j, a_i^{j'}) \mid a^j a^{j'} \in H\}$ be the copy of $\mathcal{P}$ in F_i^A.

(ii) For $i = 1, \dots, m$ let F_i^B be a copy of $F(r, k, (k-3)/2, H)$, and let V_i^B, E_i^B, and $\mathcal{P}_i^B$ be analagous to the definitions in (i).
Let $\mathcal{P}^* = \bigcup_{i=\pm 1, \dots, \pm m} \mathcal{P}_i^A \cup \bigcup_{i=1,\dots,m} \mathcal{P}_i^B$.

(iii) For $i = 1, \dots, m$ and $\ell = 1, \dots, 2r+1$ connect $e_i^\ell \in E_i^A$ and $e_{-i}^\ell \in E_{-i}^A$ with a copy S^i of $S^+(r, C_k, e, f)$.

(iv) For $i = 1, \dots, m$ join the edgesets E_i^A and E_{i+1}^A (indices mod m), with a copy M_i^A of $M(r,k)$, as described in Remark 3.4. Similarily, for $i = 1, \dots, m$ join the edgesets E_i^B and E_{i+1}^B with a copy of M_i^B of $M(r,k)$.

(v) For $i, i' = 1, \dots, m$, and $j = 1, \dots, h$, put an edge from $b_{i'}^j$ to
- a_i^j if $i \in Q_{i'}$, and
- a_i^{-j} if $i \in [m] \setminus Q_{i'}$.

Let E^* denote the set of all such edges.

Note that the graph $F(r, k, k', H)$ doesn't exist for $k' = 1$. When $k = 5$, we would need such a graph in step (ii) of the construction. This is why we cannot do Construction 3.6, or our proof of Theorem 1.1, for $k = 5$.

Proposition 3.7. *The graph $G = G(\mathcal{Q}, k, r, m)$ constructed above has the following properties.*

(i) *For some constant $C(r,k)$ independent of m, G has at most $m \cdot C(r,k)$ vertices.*

(ii) *G is r-ramsey for the cycle C_k.*

(iii) *For any of the $(r+1) \cdot m^2$ edges $e \in E^*$ introduced in step (v) of Construction 3.6, the graph $G - e$ has a C_k-free r-colouring.*

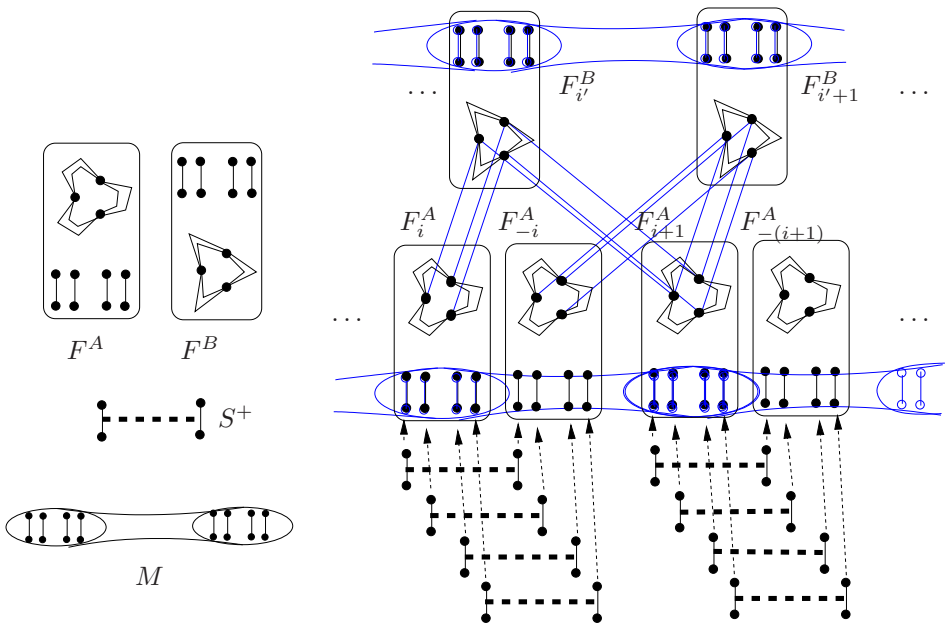

FIGURE 2. $G(\mathcal{Q}, 7, 2, m)$ where $i, i+1 \in Q_{i'}, i+1 \in Q_{i'+1}$, and $i \notin Q_{i'+1}$.

Proof.
(i) It is easily verified that

$$C(r,k) = 2|V(F_i^A)| + |V(F_i^B)| + (2r+1) \cdot |S^+| + 2|V(M)|$$

is such a constant.

(ii) Let χ be an r-colouring of the edges of G. We may assume that χ induces a C_k-free r-colouring on all copies of S^+, and M. We will show that this forces a monochromatic C_k.

Indeed, the m copies $E_1^A, \ldots, E_m^A$ of $E = \{e^1, \ldots, e^{2r+1}\}$ are connected in a cycle by copies of $M = M(r,k)$. The properties of M then imply that χ restricted to two consecutive copies of $E = W'$ must be different colourings of the set $\{\nu_1, \nu_2, \nu_3\}$. Since m is odd, there must then be some i such that χ restricted to E_i^A is the mapping ν_1. By the same argument there must be some i' such that χ restricted to $E_{i'}^B$ is ν_1.

Now i is either in $Q_{i'}$ or it isn't.

Claim 3.8. *If $i \in Q_{i'}$ then there is a monochromatic C_k consisting of vertices in $V_i^A \subset V(F_i^A)$ and $V_{i'}^B \subset V(F_{i'}^B)$.*

Proof. Assume that $i \in Q_{i'}$. Recall that the V_i^A are the vertices of an $r+1$-critical graph H. Thus the map $\chi' : V_i^A \to [r]$, on the induced from χ by $\chi'(a_i^j) = \chi(a_i^j b_i^j)$, must assign the same colour to some pair of vertices that are adjacent in H. Assume, wlog, that $\chi'(a_i^1) = \chi'(a_i^2) = 1$, and that a_i^1 and a_i^2 are adjacent in H. This implies that $\chi(a_i^1 b_{i'}^1) = \chi(a_i^2 b_{i'}^2) = 1$, and since $\chi|E_i^A$ is ν_1, it also implies by Property (ii) of Proposition 3.2, that there is a $(k-1)/2$-path of colour 1 in F_i^A with endpoints

a_i^1 and a_i^2. Similarily there is a $(k-3)/2$-path in $F_{i'}^B$ of colour 1 with endpoints $b_{i'}^1$ and $b_{i'}^2$. Putting all these things together, we get a C_k of colour 1. □

On the other hand, since for $\ell = 1, \dots, 2r+1$, the edge e_i^ℓ is connected to the edge e_{-i}^ℓ by a copy of S^+ which is C_k free under χ, χ is the same whether restricted to E_i^A or E_{-i}^A. Thus the same arguments proves

Claim 3.9. *If $i \notin Q_{i'}$ then there is a monochromatic C_k consisting of vertices in $V_{-i}^A \subset V(F_{-i}^A)$ and $V_{i'}^B \subset V(F_{i'}^B)$.*

Thus whether or not $i \in Q_{i'}$, we have a monochromatic copy of C_k.

(iii) Our first task is to show that

Claim 3.10. *Any copy C of C_k in the graph G is of one of the following types.*

(I) *C is within one of the copies of $M(r,k)$ or S^+ in G.*
(II) *For some choice of $i \in [m]$, $j \neq j' \in [h]$ and $\ell \neq \ell' \in [r]$, C is made up of of $P_{(k-1)/2}^\ell(a_i^j, a_i^{j'})$, $a_i^{j'} b_{i'}^{j'}$, $P_{(k-3)/2}^{\ell'}(b_{i'}^{j'}, b_{i'}^j)$, and $b_{i'}^j a_i^j$.*

Proof. Observe that by choice of H in Construction 3.6, Proposition 2.4, and property (i) of Proposition 3.2, and there are no cycles of length k or less, except those of type (I), in the copies of F^A and F^B. Thus there are no k-cycles except those of type (I) introduced in steps (i) and (ii) of Construction 3.6.

Propositions 2.4 and 3.5 give us that the only k-cycles introduced in steps (iii) and (iv) of Construction 3.6 are of type (I), and that any cycles that contain any of the edges E^* introduced in step (v), can only contain edges of E^* and of $\mathcal{P}^*$. We have now just to show that any k-cycle introduced in step (v), i.e., any k-cycle using edges of E^*, is of type (II).

Observe that the edges in E^* actually make up h different, non-connected, bipartite graphs D^j for $j = 1, \dots, h$, where the vertex sets of D^j are $A^j = \{a_{\pm 1}^j, \dots, a_{\pm m}^j\}$ and $B^j = \{b_1^j, \dots, b_m^j\}$. There can be no odd cycles using just the edges E^* of these bipartite graphs. Thus a k-cycle C that contains an edge of E^* has to contain some path in $\mathcal{P}^*$. Infact, it has to contain at least 2 such paths from different copies of F^A or F^B. If both are from copies of F^B, ie. are $(k-3)/2$-paths, or both are from copies of F^A, then the cycle C has to contain at least 4 edges of E^*, which in both cases gives it more than k edges. If one is from a copy of F^A while one is from a copy of F^B, then to have length k, C has to contain exactly 2 edges of E^*, and so must be of type (II). □

This proves the claim. To finish proving property (iii) of the proposition, we must consider removing any edge $e = a_i^j b_{i'}^j$ where $i \in \{\pm 1, \dots, \pm m\}$, $i' \in \{1, \dots, m\}$, and $j \in \{1, \dots, h\}$, (provided the edge exists) and show that the remaining graph $G - e$ has a C_k-free r-colouring χ. We assume that $e = a_1^1 b_1^1$. This assumption loses no generality except in the sign of i, and since F_i^A and F_{-i}^A are connected by copies of S^+, the edges relevent to our agrument will get the same colour under any C_k-free r-colouring. Thus the argument for the case i is easily seen to work for $-i$. (Implicit in the assumption that $e = a_1^1 b_1^1$ exists, is the assumption that $1 \in Q_1$.)

Define $\chi : E(G) \setminus \{e\} \to [r]$ as follows.

(i) For $i = 1, \dots, m$ let

$$\chi|_{E_i^A}, \chi|_{E_{-i}^A}, \text{ and } \chi|_{E_i^B} = \begin{cases} \nu_1 & \text{if } i = 1 \\ \nu_2 & \text{if } i \neq 1 \text{ is odd} \\ \nu_3 & \text{if } i \neq 1 \text{ is even.} \end{cases}$$

These edges are all independent, so clearly contain no monochromatic copy of C_k.

(ii) By our choice of H and by Proposition 3.2 (i), χ can be extended, for $i = 1, \dots, m$, to a C_k-free r-colouring of F_i^A, F_{-i}^A, and F_i^B. Furthermore, by Proposition 3.2 (ii), the only paths in $\mathcal{P}^*$ that are monochromatic under such a colouring χ are those in $\mathcal{P}_1^A$, $\mathcal{P}_{-1}^A$ and $\mathcal{P}_1^B$.

(iii) For $i = 1, \dots, m$, χ restricts to different members of Γ on the sets E_i^A and E_{i+1}^A, so by Construction 3.3, can be extended to a C_k-free r-colouring of M_i^A. It can similarily be extended to a C_k-free r-colouring of M_i^B

(iv) For $i = 1, \dots, m$ and $\ell = 1, \dots, r$, $\chi(e_i^\ell) = \chi(e_{-i}^\ell)$, so χ can be extended to a C_k-free colouring of the copy of S^+ connecting these edges.

(v) For all edges of E^* except those between F_1^A and F_1^B, χ can take any value. Indeed, in the claim it is observed that the only copies of C_k that these edges are in are of type (II), and in step (ii) above it is observed that paths included in this type of cycle already have more than one colour.

(vi) Since H is $k+1$-critical, there exists a proper k-vertex-colouring α of the graph $H - a_1^1$ with vertices $a_1^2, \dots, a_1^h$. Define χ on the edges from F_1^A to F_1^B as follows. For all $j = 1, \dots, h$, let $\chi(a_1^j b_1^j) = \alpha(a_1^j)$. Since α is a k-vertex-colouring of H, no two vertices of V_1^A that are connected with a path of $\mathcal{P}$ have edges to F_1^B that are of the same colour. Thus none of the k-cycles of type (II) between F_1^A and F_1^B are monochromatic.

This completes the proof of part (iii) and so completes the proof of the proposition.

□

The proof of Theorem 1.1 now proceeds as follows.

Proof. Let $r \geq 2$ and odd $k \geq 7$ be integers. Let, C be $C(r,k)$ of Proposition 3.7 (i), and let $c = c(r,k) = \frac{18}{25C^2}$. Let n_0 be greater than $3C$, and large enough that

$$\frac{2^{2cn^2}}{n!} > 2^{cn^2}.$$

Given $n \geq n_0$, let m be the maximum odd integer for which $mC \leq n$. Since $n > n_0 \geq 3C$, this implies that

$$m \geq \frac{3}{5C} n.$$

For any of the 2^{m^2} different choices of

$$\mathcal{Q} = \{Q_i \subseteq [m] | i = 1, \dots, m\},$$

Proposition 3.7 (ii) gives that the graph $G(\mathcal{Q})$ of Construction 3.6 is r-ramsey for C_k. Moreover, by Proposition 3.7 (iii), any r-ramsey-minimal subgraphs of $G(\mathcal{Q})$ and $G(\mathcal{Q}')$, for different $\mathcal{Q}$ and $\mathcal{Q}'$, are distinct. Thus we have

$$2^{m^2} \geq 2^{(\frac{3}{5C})^2 n^2} = 2^{2cn^2}$$

different graphs on less than $mC < n$ vertices that are r-ramsey-minimal for C_k. The automorphism class of any such graph is at most $n!$ so there are at least

$$\frac{2^{2cn^2}}{n!} > 2^{cn^2}$$

non-isomorphic ones.

This completes the proof of the theorem.

□

4. Senders for Odd Cycles

In this section we prove Lemma 2.2, i.e we prove the existence of negative and positive signal senders $S^-(r, C_k, e, f)$ and $S^+(r, C_k, e, f)$ for cycles C_k, where $k \geq 4$.

We begin with some definitions for ℓ-graphs (ℓ-uniform hypergraphs).

Definition 4.1. A *circuit* of length h in an ℓ-graph $\mathcal{H}$ is a subgraph $\{e_1, \ldots, e_h\} \subset \mathcal{H}$ such that

$$|\bigcup_{i=1}^{h} e_i| \leq (\ell - 1)h.$$

The *girth* of an ℓ-graph is the length of its shortest circuit.

Note that this is one of several different notions of a circuit in an ℓ-graph.

Definition 4.2. An ℓ-graph is $(r+1)$-*chromatic* if $r+1$ is the minimum number of colours needed to colour its vertices such that there is no edge with all vertices having the same colour.

We will need the following technical definitions and lemma from [18].

Definition 4.3. Let $r, \ell \geq 2$ be integers.

(i) An *oriented ℓ-graph* $\overrightarrow{\mathcal{H}}$ on vertices $V = V(\overrightarrow{\mathcal{H}})$ is a set of ordered ℓ element subsets, or arcs, of V.

(ii) An (r, ℓ)-*pattern* is an ℓ-tuple of not necessarily distinct elements from $[r]$, (i.e. an element of $\prod_{i=1}^{\ell}[r]$.)

(iii) Given a set $\mathcal{C} \subset \prod_{i=1}^{\ell}[r]$ of forbidden (r, ℓ)-patterns, an r-colouring $\chi : V(\overrightarrow{\mathcal{H}}) \to [r]$ of the vertices of $\overrightarrow{\mathcal{H}}$ is $\mathcal{C}$-*excluded* if $(\chi(v_1), \ldots, \chi(v_\ell)) \notin \mathcal{C}$ for any $(v_1, \ldots, v_\ell) \in \overrightarrow{\mathcal{H}}$.

Lemma 4.4. [18] *Given integers $k, r, \ell \geq 2$, let $\mathcal{C}$ be a proper subset of the set of all (r, ℓ)-patterns, contains the monochromatic pattern $(i, i, \ldots, i)$ for $i = 1, \ldots, r$. Then there exists an oriented l-graph $\overrightarrow{\mathcal{H}} = \overrightarrow{\mathcal{H}}(\mathcal{C}, r, \ell, k)$ with girth $\geq k$, having distinguished vertices u and u', such that*

(i) *$\overrightarrow{\mathcal{H}}$ has a $\mathcal{C}$-excluded colouring, and*

(ii) *under any $\mathcal{C}$-excluded colouring, u and u' get different colours.*

We are now ready to prove Lemma 2.2. The proof is very similar to the proof of senders for 3-connected graphs in [18]. As such we are very brief in places where the proof closely follows [18].

Proof. Let $r \geq 2$ and $k \geq 4$ be integers.

Reduction to a Simpler Sender.

We will prove the existence of slightly weaker version of a negative signal sender. In particular we prove the existence of $S' = S'(r, C_k, e, f)$ which meets properties (i) and (ii) of the negative signal sender in Definition 2.1, but in which the interface egdes e and f form an induced 2-path, rather than having property (iii).

Assuming this, and observing only that when we join two copies of S' by identifying their copies of e or f we introduce no new cycles of length k or less, the lemma follows exactly as in [18]. Indeed to get a positive sender S'' in which e and f form an induced 2-path we start with a graph $G = K_{r+1} - v_1v_2$ and a vertex $\mathbf{x}$. For every vertex v in G we let $e_v = \mathbf{x}v$ be an edge of S'' and for every edge vv' in G we join the edges e_v and $e_{v'}$ of S'' with a copy of S'. This graph, S'', has the property that under any C_k-free r-colouring, the interface edges $\mathbf{x}v_1$ and $\mathbf{x}v_2$ get the same colours. Now gluing together $k+1$ copies of S'' together in a path by identifying interface edges of consecutive copies, yields a positive signal sender in which the interface edges are distance $k+1$ apart. Gluing together k copies of S'' and one copy of S' in the same manner gives a negative signal sender in which the interfaces edges are distance $k+1$ apart.

We thus have to prove the existence of $S' = S'(r, C_k, e, f)$ meeting properties (i) and (ii) of Definition 2.1, and having the property that the interface edges e and f form an induced 2-path. This takes the remainder of the section. Before we can construct S' we must define a graph F and an oriented ℓ-graph which we use in the construction.

The Graph F.

Let F' be a graph with girth k that is r-ramsey-minimal with respect to C_k, and let F be F' with an edge xy removed. (Such graphs F' exist by [17].) Note that under any C_k-free r-colouring of the edges of F, x has incident edges of every colour $(*)$. Indeed, each vertex of C_k has degree 2, so if χ were a C_k-free r-colouring of F' and x has no incident edge of colour i, then we could extend χ to a C_k-free r-colouring of F' by setting $\chi(xy) = i$. This would be a contradiction.

The Oriented ℓ-graph $\overrightarrow{\mathcal{H}}$. Let ℓ be the degree of the vertex x in F. Fix an orientation $(x_1, \ldots, x_\ell)$ of the neighborhood of x in F. Let $\mathcal{C}$ be the set of (r, ℓ)-patterns $(r_1, \ldots, r_\ell)$ such that the following holds:

> The r-colouring χ of the edges $(xx_1, \ldots, xx_\ell)$ of F defined by $\chi(xx_i) = r_i$, for all i, can not be extended to a C_k-free r-colouring of the edges of F.

By statement $(*)$, $\mathcal{C}$ contains all (r, ℓ)-patterns with less than r colours, in particular, it contains all monochromatic (r, ℓ)-patterns, so we can apply Lemma 4.4. Let $\overrightarrow{\mathcal{H}} = \overrightarrow{\mathcal{H}}(\mathcal{C}, r, \ell, k)$ be from Lemma 4.4.

Construct graph S' as follows.

Construction 4.5.

(i) For every arc $\overrightarrow{a} = (v_1, \ldots, v_\ell)$ in $\overrightarrow{\mathcal{H}}$ let $F^{\overrightarrow{a}}$ be a copy of F. The graph S is the union of these copies of F in which all vertices are distinct unless explicitly identified below. For every vertex v in $V(F)$ let $v^{\overrightarrow{a}}$ be the copy of v in $V(F^{\overrightarrow{a}})$; thus the copy $x^{\overrightarrow{a}}$ of x in $V(F^{\overrightarrow{a}})$ will have the neighbourhood $\{x_1^{\overrightarrow{a}}, \ldots, x_\ell^{\overrightarrow{a}}\}$ in $V(F^{\overrightarrow{a}})$.

(ii) For every arc $\overrightarrow{a} = (v_1, \ldots, v_\ell)$ in $\overrightarrow{\mathcal{H}}$ and every $i = 1, \ldots, \ell$, identify the vertex $x_i^{\overrightarrow{a}}$ of $F^{\overrightarrow{a}}$ with v_i of $\overrightarrow{\mathcal{H}}$.

(iii) Identify the vertices $x^{\overrightarrow{a}}$ for all $\overrightarrow{a} \in \overrightarrow{\mathcal{H}}$ and call the resulting vertex $\mathbf{x}$. As a result of this identification, we have that $N(\mathbf{x}) = V(\overrightarrow{\mathcal{H}})$.

(iv) Let $e = \mathbf{x}u$ and $f = \mathbf{x}u'$ where u and u' are the distinguished vertices of $\overrightarrow{\mathcal{H}}$ that get different colours under any $\mathcal{C}$-excluded colouring.

We will now prove that the graph S', thus constructed, is the graph $S'(r, C_k, e, f)$ that we are looking for. By the construction, it is clear that the edges e and f form an induced 2-path.

We begin by proving the following claim.

Claim 4.6. *Let C_* be a cycle in S' of length k or less. Then C_* is entirely within $F^{\overrightarrow{a}}$ for some $\overrightarrow{a} \in \overrightarrow{\mathcal{H}}$.*

Proof. Let C_* be a cycle in S' of length $r \leq k$. Since F has girth $k \geq 4$, there are no edges among the neighbours of $\mathbf{x}$, thus C_* contains some vertex v not in $\{\mathbf{x}\} \cup N(\mathbf{x})$. Assume, wlog, that $v \in V(F^{\overrightarrow{a}}) - (\{\mathbf{x}\} \cup N_{F^{\overrightarrow{a}}}(\mathbf{x}))$. If $V(C_*) \subset V(F^{\overrightarrow{a}})$, then we are done, so assume this isn't true. Then because the only vertices of $F^{\overrightarrow{a}}$ that are adjacent to vertices outside of $F^{\overrightarrow{a}}$ are in $\{\mathbf{x}\} \cup N_{F^{\overrightarrow{a}}}(\mathbf{x})$, $C_* \cap F^{\overrightarrow{a}}$ must contain a path P between two such vertices, which contains v. If the two vertices are $\mathbf{x}$ and some neighbour, then because $F^{\overrightarrow{a}}$ has girth k, the aforementioned path P has length at least k, and so C_* is entirely within $F^{\overrightarrow{a}}$. If the two vertices are in $N_{F^{\overrightarrow{a}}}(\mathbf{x})$, then they share the neighbour $\mathbf{x}$, so by the girth of $F^{\overrightarrow{a}}$, the path P has length at least $k-1$, but the only common neighbour of the two vertices would have to be within $F^{\overrightarrow{a}}$, so again, C_* is entirely within $F^{\overrightarrow{a}}$. □

The following two claims verify properties (i) and (ii) of Defintion 2.1.

Claim 4.7. *There exists a C_k-free r-colouring of S'.*

Proof. Let χ be a $\mathcal{C}$-excluded r-colouring of $\overrightarrow{\mathcal{H}}$. We define a colouring χ' on the edges of S' as follows, beginning with the edges incident to $\mathbf{x}$. For any vertex v in $\overrightarrow{\mathcal{H}}$, let

$$\chi'(\mathbf{x}v) = \chi(v).$$

By our choice of $\mathcal{C}$ this colouring can be (and is) extended to a C_k-free r-colouring of each copy of F. Since by the previous claim, this accounts for all copies of C_k in S', we are done. □

Claim 4.8. *Under any C_k-free r-colouring of S', edges e and f get different colours.*

Proof. Let χ' be a C_k free r-colouring of S'. Then the vertex colouring χ of $\overrightarrow{\mathcal{H}}$ defined by

$$\chi(v) = \chi'(\mathbf{x}v),$$

is a C-excluded r-colouring of $\overrightarrow{\mathcal{H}}$. Thus $\chi(u) \neq \chi(u')$, so $\chi'(e) = \chi'(\mathbf{x}u) = \chi(u) \neq \chi(u') = \chi'(\mathbf{x}u') = \chi'(f)$, as needed. □

This completes the proof of the exsitence of $S'(r, C_k, e, f)$, so by the discussion at the beginning of the proof, completes the proof of the lemma.

□

5. Concluding Remarks

By [18], Theorem 1.1 holds for $k = 3$ as well.

It is well known that the result cannot hold when k is even. Indeed, an immediate corollary of the Theorem 1.1 is similar statement that there is some other constant $c' = c'(r, k)$ such that for large enough n there exist a graph G_n on at most n vertices with at least $c'n^2$ edges that is r-ramsey-minimal for C_k. The fact that the Turan number for bipartite graphs is $o(n^2)$ shows that the 'minimal' part of this is impossible if k is even.

For cycles then, this leaves the question open only for $k = 5$: Is C_5 highly ramsey-infinte?

The existance of signal senders is a useful tool in ramsey-minimal constructions. As we stated earlier, by [7], [10] and [18], senders exist for any 3-connected graph G. At the same time, by the discussion above, no bipartite graph can be highly ramsey-infinite. Apart from being bipartite, do 3-connected graphs have any other obstruction to being highly ramsey-infinite? That is to say, is any 3-connected non-bipartite graph G highly ramsey-infinite?

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